Practical Methodology

[almost] Dr. Kaitlin Palmer
San Diego State University

J&M: 9.4.1

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Practical Methodology

- Project Design
- Performance Metrics
- Pipeline
  - Baseline Algorithm Selection
  - Optimization
- Data collection
- Hyperparameter Tunining
- Debugging

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Performance Metrics

- Consider- training data will never be perfect
- How good does the system need to be?
  - Accurate, precise?
  - Better than competitors?

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Error Types

- Accuracy
  - Rare events, set model equal to 0
- Precision/Recall
  - Fraction of correctly predicted observations
  - Fraction of true events that were detected
- Type I and II error

	H_0 True	H_0 False
Reject H_0	Type I Error	Correct Rejection
Fail to Reject H_0	Correct Decision	Type II Error

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Performance Metrics

- Precision/Recall
- F-scores (manager's like single numbers)

$$F = \frac{2pr}{p + r}$$

Performance Metrics

- Maybe we shouldn't leave everything to the machines…
- AI only useful if they can reduce the payload for human analysts
- Coverage
  - Proportion of which the machine is able to produce a response
  - Perfect accuracy- refuse to do anything

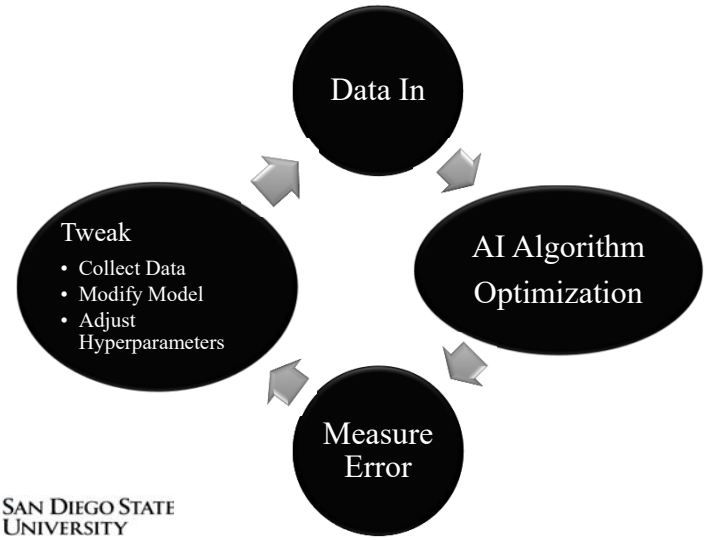
Other Performance Metrics

- Categorizing emotion?
  - User satisfaction
- Adverting
  - Sales outcomes
  - Click-through rates
- Numerous project specific metrics
  - Your turn!

Next Steps

- We have a performance metric
- Establish "end-to-end" system (ASAP)
  - Prototype
  - Check predictions, debug
  - Add complexity
  - Check predictions, debug

Pipeline



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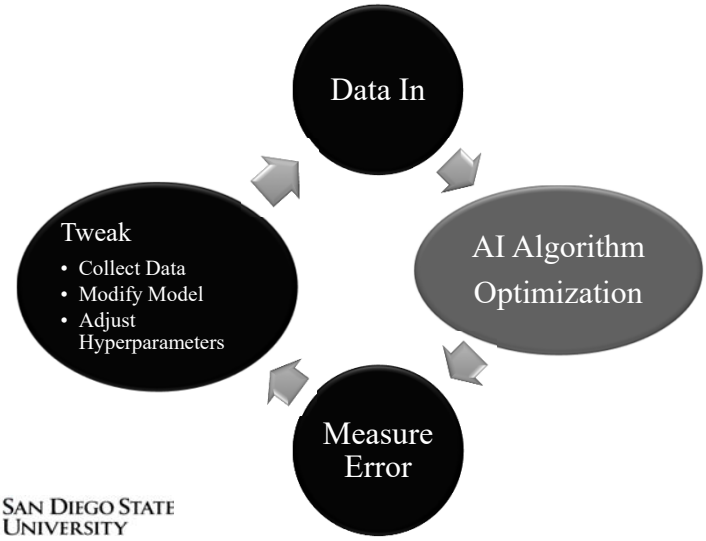
Pipeline



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Deep Learning Algorithms

- Deep learning first? Why? Why not?
- Statistics models (there are lots!)
  - GMM, PCA
  - Random forest
  - CART
  - GAMs, GLM, linear regression
- Read with a skeptical eye

Deep Learning Algorithms

- No Data Structure
  - Lots of measurements, sequence of features
  - Feed forward network (2-3 hidden layer, RELU)

Deep Learning Algorithms

- Structured Data (images)
  - Convolutional network
    - ReLU
    - Leaky ReLU
    - PreLus
    - Maxout

Deep Learning Algorithms

- Sequences (in/out)
  - Gated recurrent net
    - LTSM
      - Train with momentum or SGD
      - Clip the gradients to avoid instability in back propagation
    - Set forget gate high (remember everything)
    - GRU

Deep Learning Algorithms

- Is your problem well understood?
  - Copy a trained model (architecture)
  - Convolutional networks trained on ImageNet

Optimization Algorithm

- Stochastics Gradient Descent
  - Decaying learning rate
    - Linear
    - Exponential
    - Minimum learning rate
  - Adam (adaptive moments)
  - Introduce batch normalization quickly if optimization is problematic

Additional Setup Considerations

- How much training data do you really have?
  - If less than millions see below
- Probably need to use regularization
  - Early stopping (almost always!)
  - Dropout
  - Batch normalization (reduces generalization error), skip dropout

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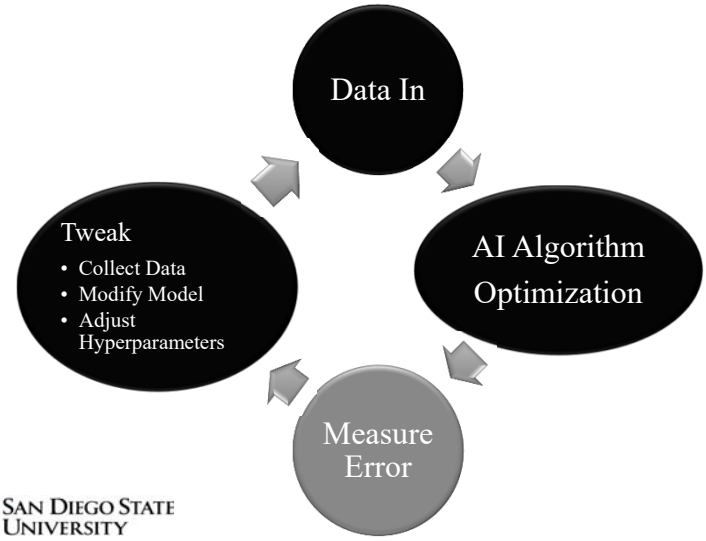
Round 1



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Measure Error

- How well did the model perform wrt desired metric
  - On the training data (poor)
  - On the test data (poor)
- Poor performance on training set will not benefit from more data
  - Add more layers
  - Add more hidden units to each layer

When to Collect More Data

- Poor performance on training set
  - Try adjusting the learning algorithm
  - Learning rate (hyperparameter)
- Still no good? Humm..
  - Might be the data

Data Problems

- Are the features rich enough for the intended outputs?
- Are the data too noisy?
- Are the data correctly labeled?
  - Unsupervised learner error
  - Human error

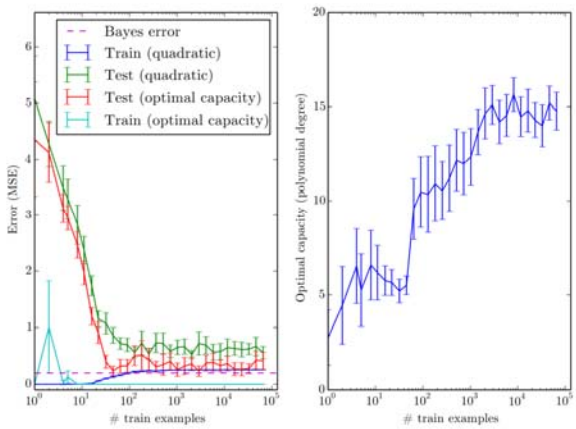
When To Collect More Data

- How well did the model perform?
  - On the training data (good)
  - On the validation data (poor)
- Poor on the test set
  - Features in test set non-representative or incomplete
  - Collect more data!

Collecting More Data

- Dataset Augmentation
- How Much?
  - How much is available?
  - No more available
    - shrink the model
- Know your system
  - Plot generalization error vs training samples
  - Usually will need orders of magnitude (logarithmic scale)

How Much to Collect



Goodfellow et al. 2016

Getting the Most Out of Your Data and Algorithm

- Understand your data
- Understand your algorithms (especially limitations)
- Monitor output and adjust experimental system

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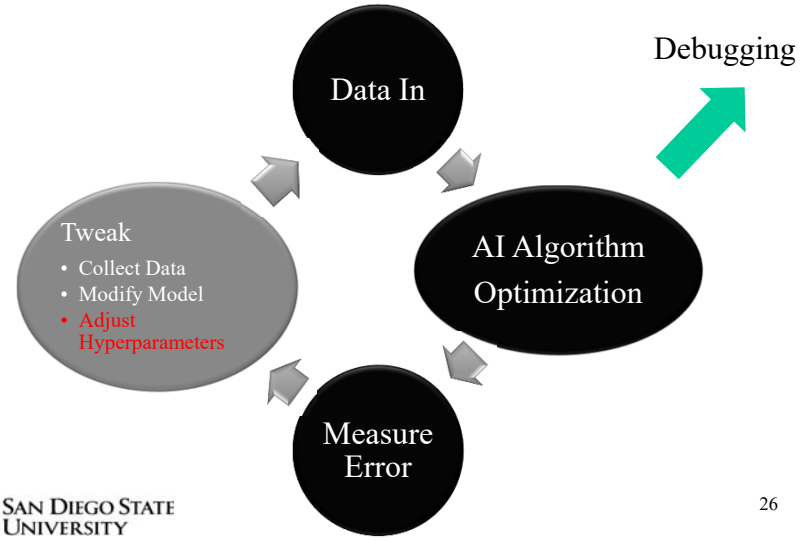
Round 1



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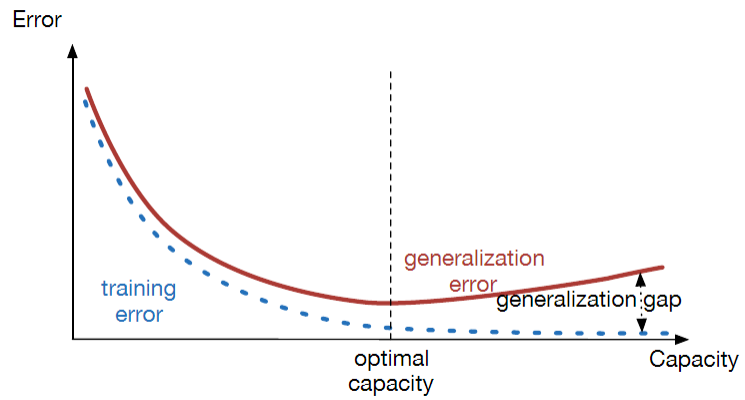
Hyperparameter Tuning

- Tradeoff between run-time and memory
- Model quality
- Can be chosen manually (extensive knowledge or saint-like patience) or programmatically (computationally costly)
- Lowest generalization error for a given memory/time allowance

Manual Tuning

- Need to match the capacity of the model with the complexity of the task
- Model capacity determined by:
  - Representational capacity of the model
  - Ability of the learning algorithm(s) to minimize cost function used to train the model
  - Degree of regularization produced by the cost and traning procedure

Model Capacity

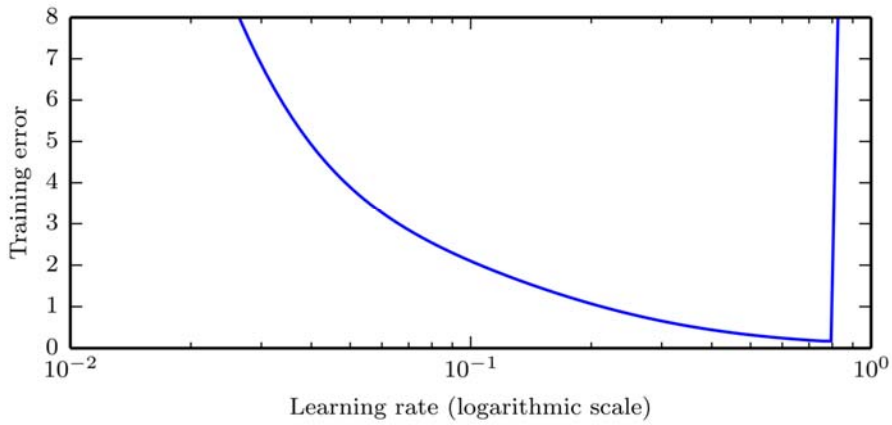


Goodfellow et al. 2016

Model Capacity

- Overfitting
  - Value of hyperparameters too large
    - Too many hidden units per layer
  - Value of hyperparameters too small
    - Weight decay coefficient near 0 greatest effective capacity of the algorithm
  - Some hyperparameters discrete
    - Units in a layer, switches

Learning Rate

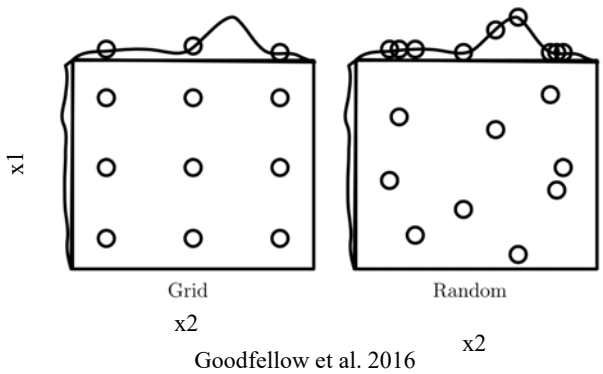


Goodfellow et al. 2016

Automatic Hyperparameter Optimization

- Preferred when
  - No prior knowledge of the filed
  - Large (>40) number of parameters available
- Develop hyperparameter optimization algorithms
  - Optimize our objective function (training error)
  - Wrap around learning algorithm

Hyperparameter Optimization

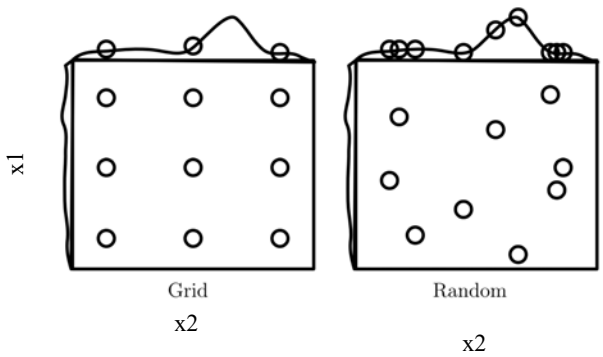


Goodfellow et al. 2016

Hyperparameter Optimization

- Grid Search
  - Three or fewer hyperparameters
    - Computational cost grows exponentially
  - "Bin" hyperparameters
  - Parameter selection on log scale
  - Monitor/update selected hyperparameters (e.g. edge of value range, scale)

Hyperparameter Optimization

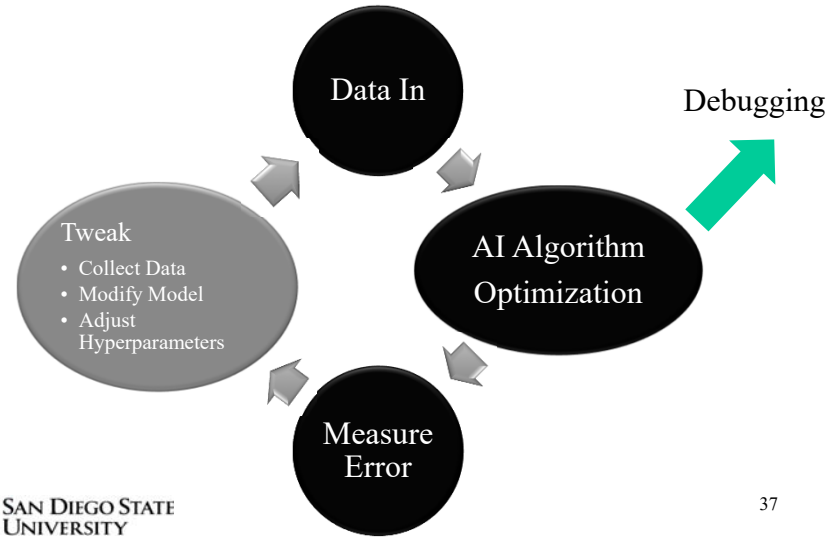


Goodfellow et al. 2016

Hyperparameter Optimization

- Random Search
  - Less computationally expensive (independent exploration of the hyperparameters)
  - Define marginal distribution for each hyperparameter (Bern, binary, uniform, etc.)
  - Sample log number of hidden units form the distribution
  - Monitor/update selected hyperparameters (e.g. edge of value range, scale)

Round 1



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Debugging

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Debugging

- Known unknowns
  - Programmatic errors
- Unknown unknowns
  - If we didn't know what to expect, how can we tell if the result is good? Is 10% test error good?
  - Adaptive (that's why we use deep learning)

Debugging

- Visualize / check output
  - Superimpose images for classification
  - Listen to generative
- Visualize mistakes-
  - Identify training errors in the data consistent across all datasets

Debugging

- High training error
  - Underfitting or training error
  - Small samples can be fit by small models
  - If not….

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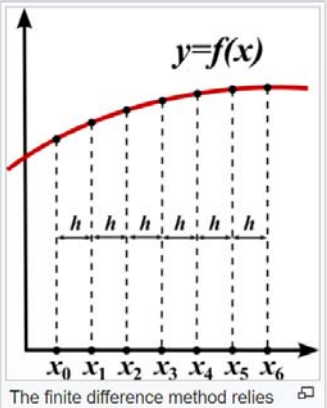
Debugging

- Compare b merical derivatives
  - Some sys lient decent
  - Implemer
  - Use finite l differences (calculus



The finite difference method relies on discretizing a function on a grid.

Wikipedia

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Debugging

- Visualize!
  - Monitor histograms of activations and gradients
  - Are units saturated
  - How often are rectifiers off?
  - Magnitude of parameter gradients to the magnitude of parameters themselves- have some feature parameters stalled?

Summary

- Know your question
  - Key error metrics
  - Coverage
- Know your data
  - Where might there be errors or inconsistencies?
- Know your algorithm
- Visualize, VISUALIZE, VISUALIZE